Teacher's Signature: .....

Experiment No. 2

Dated : 20/4/25

Objective : To find zeroes of a quadratic polynomial geometrically

Materials Required:

white paper, graph paper, geometry box, a scale and fevistick.

Pre - Requisite Knowledge (with theory)

Quadratic polynomial

A polynomial of the form ax^2+bx+c, $a \neq 0$ is called a quadratic polynomial.

A real number is said to be a zero of a polynomial $p(x) = ax^2+bx+c$, $a \neq 0$, if $p(x) = 0$ i.e. $ax^2+bx+c = 0$ (the value of the polynomial vanishes). Here x is also called the solution of equation $ax^2+bx+c = 0$. Any quadratic polynomial can have at most two zeroes.

Graph of a quadratic polynomial

The graph of any quadratic polynomial ax^2+bx+c, $a \neq 0$ has parabolic shapes (either ~~of the~~ open upward or open download depending on whether $a>0$ or $a<0$)

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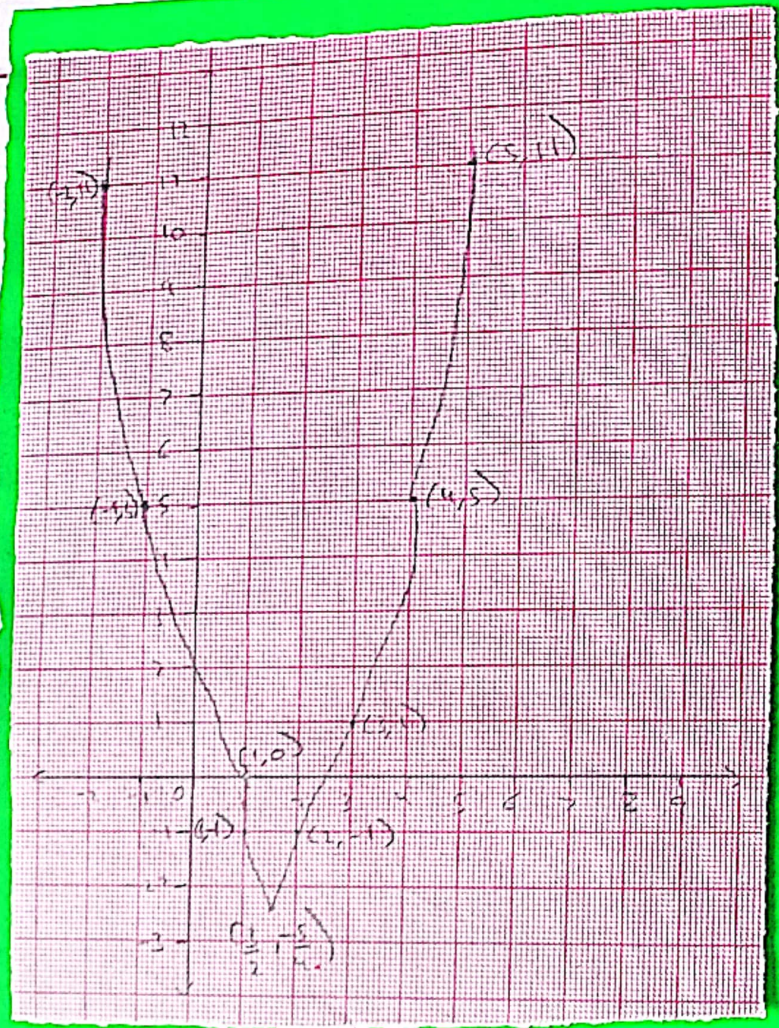

x	0	1	-1	2	3	4	5	-2	$\frac{3}{2}$
y	1	-1	5	-1	1	5	11	11	$\frac{-5}{4}$
(x,y)	(0,1)	(1,-1)	(-1,5)	(2,-1)	(3,1)	(4,5)	(5,11)	(-2,11)	$(\frac{3}{2}, \frac{-5}{4})$

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Procedure :

1. Consider a quadratic polynomial $x^2 - 3x + 1$
2. Let $x^2 - 3x + 1$
3. obtain the tables values of x and y

Case I :

$a > 0$ $a < 0$

No. of zeroes = 2 (Different)

Case II :

$a > 0$ $a < 0$

No. of zeroes (equal zeroes

Case III :

$a > 0$ $a < 0$.

No. of zeroes - Null

Result - quadratic polynomial can have atmost 2 zeroes.

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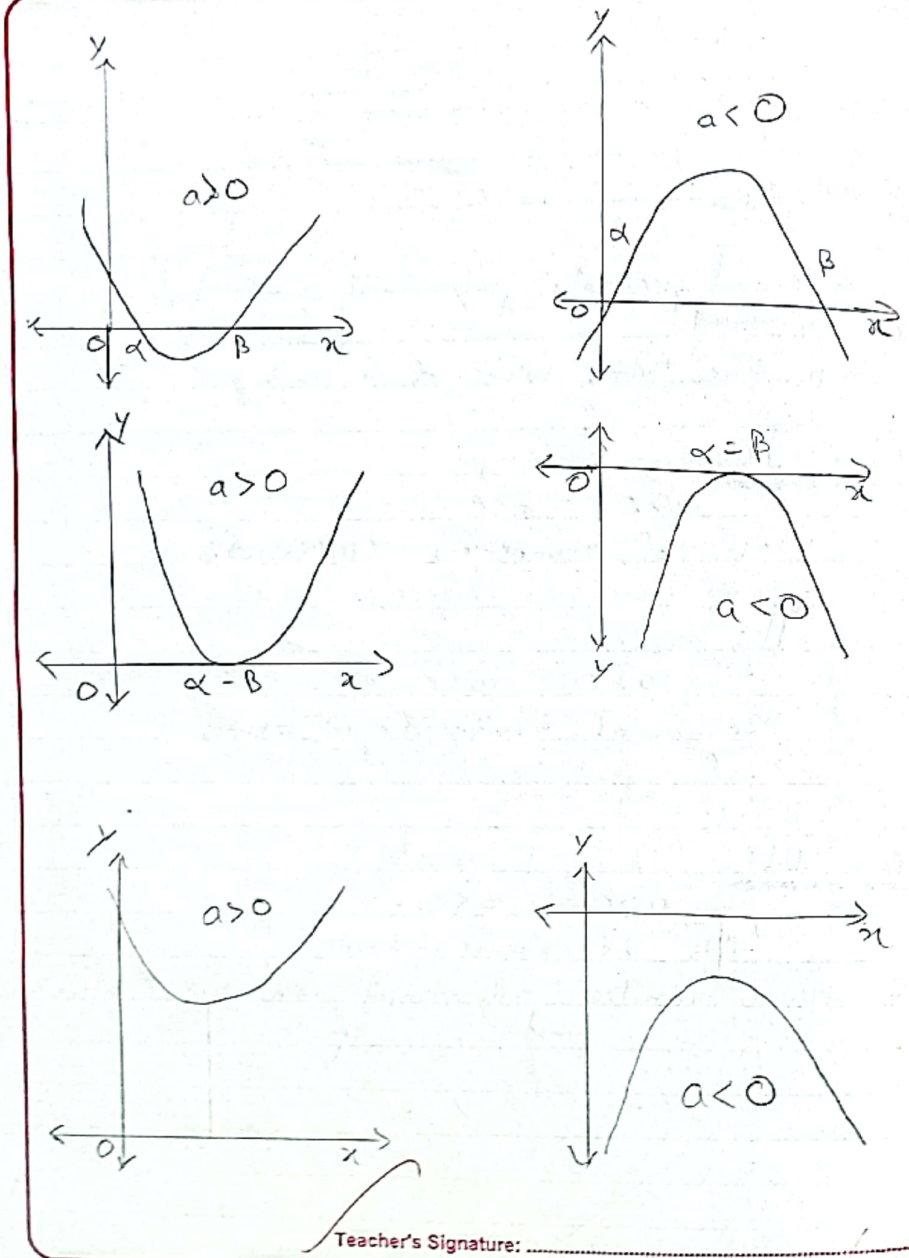

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Case IV:

Sum of zeroes of a quadratic polynomial ax^2+bx+c, $a \neq 0$ is

$$\alpha+\beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$$

Case V:

Products of zeroes, $\alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$